The Absolute Basics (Read This First)

VLA Robot Project — Foundations

Robotics + AI Mentor Notes

August 13, 2026

Abstract

No setup, no commands. Just plain-language foundations so nothing later feels like magic. Mental model of the whole project: **Body (Isaac Sim), Nerves (ROS 2), Brain (our code → VLA).**

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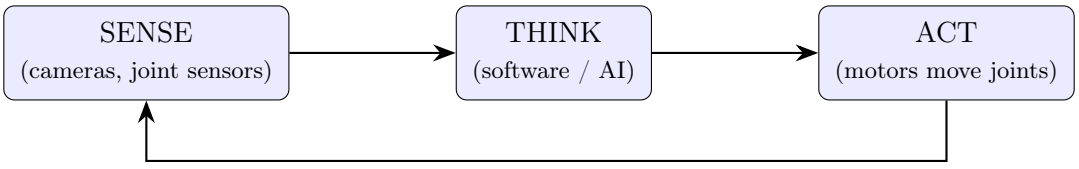
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1 What is a robot (in our context)?

A robot is a machine that **senses, thinks, and acts**:



Our robot is a **robot arm** (a manipulator), like a human arm bolted to a table:

- **Links** = rigid segments (upper arm, forearm).
- **Joints** = movable connections (shoulder, elbow, wrist).

- **End-effector** = the "hand" / gripper at the tip.

  Robot "moving" = **motors rotate the joints**, repositioning the hand in space.

2 What does "control a robot" actually mean?

At the lowest level, controlling a robot = **choosing the angle of each joint**.

A 6-joint arm has a joint-angle vector $\mathbf{q} \in \mathbb{R}^6$, for example:

$$\mathbf{q} = \begin{bmatrix} q_1 & q_2 & q_3 & q_4 & q_5 & q_6 \end{bmatrix}^\top = \begin{bmatrix} 0^\circ & -45^\circ & 90^\circ & 0^\circ & 45^\circ & 0^\circ \end{bmatrix}^\top.$$

Change those numbers $\Rightarrow$ the arm moves. **Every robot motion is ultimately a change in this vector q.**

3 Why simulation? What is Isaac Sim?

Real robot arms are expensive, slow, and breakable. So we practice in a **simulator** — a physics-accurate virtual world.

NVIDIA Isaac Sim gives us:

- A virtual robot arm.
- Virtual cameras producing real-looking images.
- Virtual objects (red cube, box).
- Real physics (gravity, collisions, friction).

  What we learn in sim mostly transfers to real robots later — **Phase 17, "sim-to-real."**

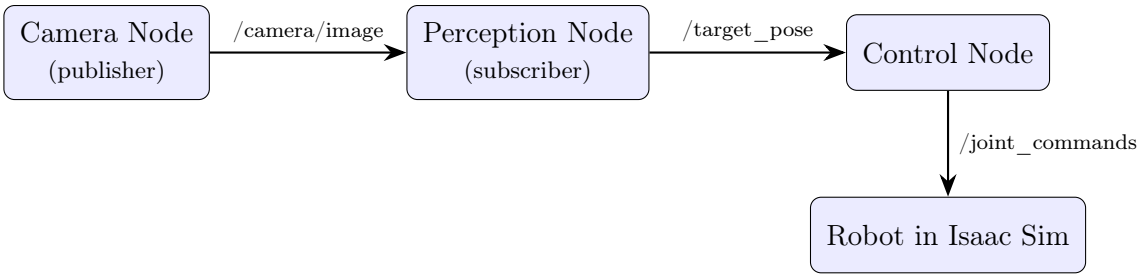
4 What is ROS 2?

The robot's software is **many small programs that must talk to each other**: one reads the camera, one detects objects, one decides the action, one drives the motors.

ROS 2 (Robot Operating System 2) is the "postal service" connecting them. Not an OS like Windows — a **communication + tooling framework** for robots.

Core vocabulary (deep dive in Phase 3):

- **Node** = one small program (e.g., "camera reader").
- **Topic** = a named channel messages flow through (e.g., `/camera/image`).
- **Publisher** = a node that sends messages onto a topic.
- **Subscriber** = a node that listens to a topic.
- **Message** = the data itself (an image, a joint command, etc.).



5 What is a VLA — in one sentence?

- A **VLM** (Vision-Language Model): image + text → **describes** ("A red cube is on the table.").
- A **VLA** (Vision-Language-Action model): image + instruction → **acts** (outputs *numbers that move the robot*).

VLA = VLM + the ability to produce robot actions.
Turning understanding into motion is what makes robotics hard and interesting.

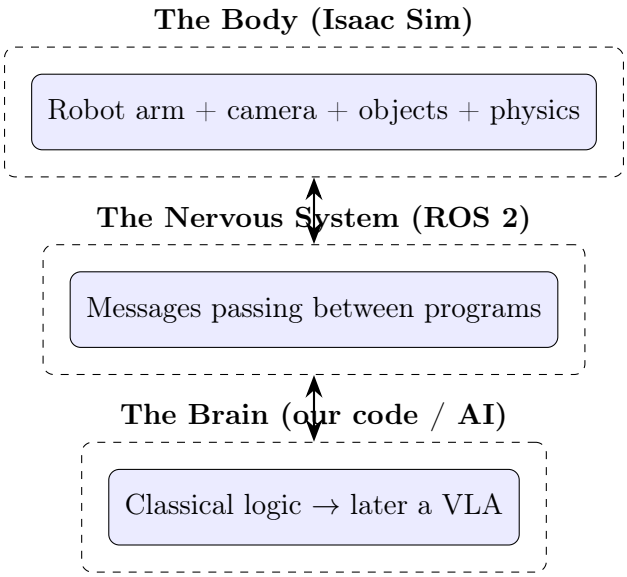
6 How the pieces fit the final goal

Goal: *"Pick up the red object and put it in the box."*

1. **Sense** — cameras (Isaac Sim) send images via **ROS 2 topics**.
2. **Think** — software finds the red object and decides where to move.
3. **Act** — send joint-angle commands to the robot via ROS 2.
4. **Repeat** — look again, adjust (feedback loop).

We first build step 2 with **classical hand-written logic** (Phase 7), then later **replace the brain with a learned VLA** (Phases 9–16).

7 The mental model to hold onto



- **Isaac Sim** = the body + world.
- **ROS 2** = the nervous system carrying signals.
- **Our code (eventually a VLA)** = the brain.

Three words: **Body, Nerves, Brain.**

Next step

Once these basics feel clear, we return to **Phase 0**: verify the environment (see `phase0_environment.md`).